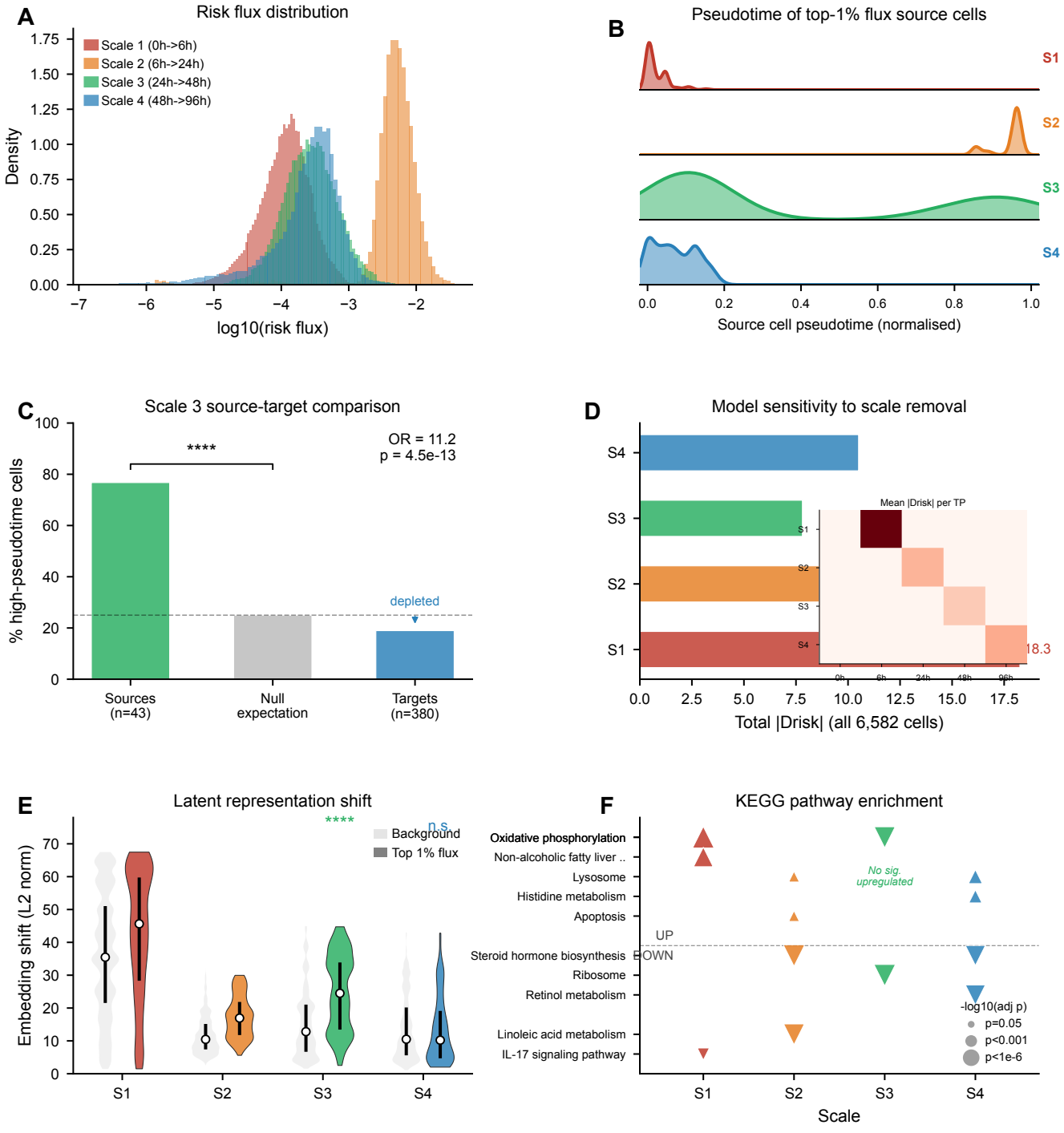

A
Risk flux distribution
Scale 1 (0h->6h)
Scale 2 (6h->24h)
Scale 3 (24h->48h)
Scale 4 (48h->96h)
Density
log10(risk flux)
B
Pseudotime of top-1% flux source cells
S1
S2
S3
S4
Source cell pseudotime (normalised)
C
Scale 3 source-target comparison
OR = 11.2
p = 4.5e-13
****
depleted
% high-pseudotime cells
Sources (n=43)
Null expectation
Targets (n=380)
D
Model sensitivity to scale removal
Mean |Drisk| per TP
18.3
Total |Drisk| (all 6,582 cells)
E
Latent representation shift
****
n.s.
Background
Top 1% flux
Embedding shift (L2 norm)
F
KEGG pathway enrichment
Oxidative phosphorylation
Non-alcoholic fatty liver ..
Lysosome
Histidine metabolism
Apoptosis
Steroid hormone biosynthesis
Ribosome
Retinol metabolism
Linoleic acid metabolism
IL-17 signaling pathway
UP
DOWN
No sig. upregulated
-log10(adj p)
p=0.05
p<0.001
p<1e-6
Scale